

Erdős Problem #122

EP-122

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For which number theoretic functions f is it true that, for any $F(n)$ such that $f(n)/F(n) \rightarrow 0$ for almost all n , there are infinitely many x such that $\frac{\#\{n \in \mathbb{N} : n + f(n) \in (x, x + F(x))\}}{F(x)} \rightarrow \infty$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>